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Modelling Economic Policy Issues

Do investment provisions in regional agreements foster growth? A Quasi-experimental analysis of the ASEAN–Japan partnership

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ABSTRACT

Regional economic agreements have increasingly shifted from tariff liberalization to investment-related institutional coordination, yet the macroeconomic effects of these provisions remain insufficiently examined. This issue is especially relevant for the Association of Southeast Asian Nations (ASEAN), which has remained resilient amid weakening global foreign direct investment flows and in which Japan has long played a central role in regional production networks. This study treats the implementation of the ASEAN–Japan Comprehensive Economic Partnership Agreement (AJCEP) as a quasi-natural institutional shock and examines whether investment-related regional cooperation promotes economic growth. Using annual panel data for nine Asia-Pacific economies from 2000 to 2023, the study applies a difference-in-differences (DID) framework with country and year fixed effects. The results show that AJCEP significantly increased gross domestic product growth in treated ASEAN economies relative to the control group. This finding remains robust after event-study tests, country-specific policy timing checks, country-clustered and wild-cluster bootstrap inference, controls for competing regional agreements, adjustments for the global financial crisis, propensity score matching DID estimation, placebo tests, and alternative sample composition checks. Auxiliary channel evidence further suggests that AJCEP is associated with higher foreign direct investment inflows, stronger production-network trade linkages, and improved regulatory quality. Heterogeneity analysis indicates that the growth effect is not clearly differentiated by income level, but remains positive across both interventionist industrial-policy and more market-oriented environments. These findings suggest that investment-related regional agreements foster growth more effectively when supported by credible domestic implementation, regulatory alignment, transparent investment procedures, institutional capacity building, and production-network upgrading.

1. Introduction

Regional economic integration in East Asia has deepened substantially in recent years, with regional trade and investment policies playing an increasingly important role in shaping international capital flows and economic growth trajectories (Argiro and Konstantinos, 2025; Caporale et al., 2021). For many developing economies, sustained growth depends not only on domestic capital accumulation (Dwivedi, 2026) but also on technological progress (Madsen and Mayandy, 2026), industrial upgrading (Zhao et al.,

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2025), and deeper integration into global production networks (Wei et al., 2025). In this context, foreign direct investment (FDI) is widely regarded as a key channel through which host economies may obtain external capital, technology transfer, managerial knowledge, and improved access to international markets (F. Wang et al., 2026). These mechanisms can enhance productivity, support industrial restructuring, and facilitate the integration of developing economies into global value chains and transnational production networks (Gavrila-Paven, 2026; Le et al., 2024). The empirical relevance of this issue is evident in recent global and regional investment patterns. According to the United Nations Conference on Trade and Development (UNCTAD, 2025), global FDI fell by 11% to USD 1.5 trillion in 2024, marking the second consecutive year of decline.

In contrast, the Association of Southeast Asian Nations (ASEAN) remained resilient as an investment destination. The ASEAN Investment Report 2025 shows that FDI inflows to ASEAN rose by about 8% to USD 226 billion in 2024, while the region maintained its position as the top FDI recipient among developing regions for the fourth consecutive year (UNCTAD, 2025). ASEAN–Japan economic relations also remain substantial: official ASEAN statistics show that total two-way trade between ASEAN and Japan reached USD 236.4 billion in 2024 (The ASEAN Secretariat, 2025). These patterns suggest that investment-related regional agreements are not merely diplomatic or legal instruments, but also form part of the institutional environment through which firms assess market access, regulatory predictability, and long-term production-network decisions.

Although the role of FDI in economic growth has been widely examined, its effects are shaped not only by the volume of capital inflows but also by the institutional and policy environment in which investment takes place (Hou and Wang, 2026; Maimone Ansaldo Patti et al., 2026). This distinction is particularly important in East Asia, where trade, investment, and production networks are closely interconnected. However, existing studies have mainly focused on the growth effects of FDI inflows or the determinants of cross-border investment, while paying less attention to whether investment-related policy changes themselves generate measurable macroeconomic effects (Ashraf et al., 2024). This leaves an important gap in understanding how regional institutional arrangements influence economic growth beyond their effects on investment flows alone.

The economies of the Regional Comprehensive Economic Partnership (RCEP[1]) provide a valuable empirical context for examining this issue (Argiro and Konstantinos, 2025). Over the past decades, Japanese firms have established extensive cross-border production networks in East Asia, particularly in the automobile, electronics, and machinery manufacturing sectors (Kumar, 2025; Peng, 2024). These networks have promoted regional trade, technology diffusion, industrial specialization, and the spatial reorganization of production activities. Alongside these developments, institutional cooperation between Japan and ASEAN[2] has progressively deepened. The ASEAN–Japan Comprehensive Economic Partnership Agreement (AJCEP[3]), signed in 2008 and implemented in stages, represents a significant institutional milestone in this process (AJCEP, 2008). By strengthening regional economic cooperation and policy coordination, AJCEP contributed to a more stable and predictable environment for cross-border investment. It therefore provides a useful empirical setting for identifying the economic effects of investment-related institutional changes in a region where production networks rely heavily on credible policy frameworks and regulatory coordination (Imai, 2020).

Based on this background, this study treats the implementation of AJCEP as a relatively exogenous investment-related institutional shock. It examines its effect on economic growth within a quasi-experimental framework. Using panel data for nine RCEP member economies from 2000 to 2023, the study applies a difference-in-differences (DID) approach to compare economic performance between treatment and control groups before and after the policy implementation. The results show that, following the implementation of AJCEP, the economic growth rate of treated ASEAN economies increased significantly relative to the control group. Event-study analysis confirms that there were no significant pre-policy differences in trends, supporting the parallel-trends assumption underlying the estimation. A series of robustness checks, including the exclusion of pandemic years, controls for competing regional agreements, country-specific policy timing, global financial crisis adjustments, propensity score matching combined with DID estimation, placebo tests, and alternative sample-composition checks, consistently support the stability of the baseline results. Additional channel-level evidence further suggests that AJCEP is associated with higher FDI inflows, stronger production-network trade linkages, and improved regulatory quality. Heterogeneity analysis indicates that the growth effect is not clearly differentiated by income level, but remains positive across both interventionist industrial-policy and more market-oriented environments. These findings suggest that domestic policy-orientation environments are more relevant to the transmission of investment-related regional agreements into growth than income-level differences in this sample.

This study contributes to the literature in three specific ways. First, it shifts the empirical focus from FDI inflows to investment-related institutional provisions in regional agreements. While the FDI–growth literature has long examined whether capital inflows promote economic performance, recent evidence shows that investment provisions embedded in preferential trade agreements can shape multinational production and firms' participation in global value chains (Miroudot and Rigo, 2022). At the same time, the existing evidence remains inconclusive as to whether international investment agreements consistently attract FDI inflows (Gopalan et al., 2023). This study extends this line of research by examining whether investment-related institutional commitments generate

[1] Regional Comprehensive Economic Partnership (RCEP): Regional Comprehensive Economic Partnership, a trade and investment agreement among 15 Asia-Pacific economies, including ASEAN members, China, Japan, South Korea, Australia, and New Zealand, promoting regional economic integration.

[2] Association of Southeast Asian Nations (ASEAN): Association of Southeast Asian Nations, a regional organization of ten Southeast Asian countries focused on economic growth, trade cooperation, and political stability.

[3] ASEAN-Japan Comprehensive Economic Partnership Agreement (AJCEP): Agreement on Comprehensive Economic Partnership among Japan and Member States of the Association of Southeast Asian Nations, signed in 2008 to enhance trade, investment, and institutional coordination between Japan and ASEAN countries.

measurable macroeconomic growth effects.

Second, the study contributes to the literature on regional agreements and production-network integration. Existing research shows that deeper trade agreements are associated with stronger participation in global value chains and cross-border production networks (Laget et al., 2020). Recent evidence from Asia-Pacific economies further shows that deeper regional trade agreements can strengthen supply-chain resilience by improving institutional quality and reducing trade costs (Yu and Hu, 2026). By focusing on AJCEP, this study links investment-related agreement provisions to the ASEAN–Japan economic relationship, where trade, investment, institutional coordination, and production-network linkages are closely embedded. This perspective helps clarify how regional institutional arrangements may influence growth through both capital-flow and production-network channels.

Third, the study provides quasi-experimental evidence from East Asia by treating the implementation of AJCEP as a policy shock and applying a DID framework. Ma (2022) similarly uses quasi-experimental variation to show that broader economic integration under the Belt and Road Initiative improved economic performance, with suggestive evidence pointing to trade and inward FDI as potential channels. This study narrows the focus to the investment-related institutional provisions embedded in a specific regional agreement and examines whether they generate measurable growth effects. In line with recent DID literature that emphasizes treatment timing, dynamic treatment patterns, and identification validity (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021), this study supplements the baseline estimates with event-study analysis, placebo tests, country-specific policy-timing checks, small-cluster inference, and additional robustness tests. This design moves the analysis beyond simple correlations between FDI and growth, providing policy-relevant evidence on whether regional investment rules can translate into domestic growth benefits.

2. Literature review, institutional background, and theoretical mechanism

2.1. Review of previous literature

2.1.1. FDI and economic growth

Although the relationship between FDI and economic growth has been extensively studied, the evidence remains inconclusive. Some research suggests that FDI can significantly promote growth in host countries through channels such as technology spillovers (Razzaq et al., 2021), capital deepening (Lahiri et al., 2024), and productivity enhancement (Ashraf et al., 2024). However, other studies emphasize that these effects are not universal and depend critically on factors including human capital (Kebede et al., 2024), financial development (Mensah et al., 2025), and the quality of the host country's institutional environment (Chinh et al., 2026). When accounting for endogeneity and country-specific heterogeneity, the direct effect of FDI on economic growth may not be robust, highlighting that the economic impact of cross-border investment is conditional on specific institutional and policy contexts.

2.1.2. Regional agreements and investment cooperation

A growing body of literature emphasizes that investment policies and institutional arrangements play a central role in shaping cross-border investment behavior. In particular, transparent investment rules can reduce regulatory uncertainty (Fauzel et al., 2025), while institutional stability shapes firms' long-term investment expectations (Chinh et al., 2026). Policy coordination also affects cross-border investment decisions by strengthening the credibility and predictability of the investment environment (M. Wang et al., 2026). Consistent with this institutional perspective, Tag and Degirmen (2022) find that FDI inflows are higher in countries with stronger rule-of-law institutions, greater trade freedom, and lower regulatory barriers to investment and business activity. In the Southeast Asian context, Kubo (2019) shows that Japanese FDI is influenced by host-market demand and the agglomeration of Japanese firms, highlighting the role of established regional production networks in investment location decisions. Reflecting these developments, regional economic agreements have evolved beyond traditional tariff reductions to encompass investment facilitation, institutional coordination, and broader forms of economic cooperation. By lowering transaction costs, enhancing policy transparency, and strengthening institutional stability, such arrangements are likely to influence both the direction of cross-border investment and its longer-term macroeconomic effects.

Building on this perspective, recent studies further highlight that FDI is increasingly embedded in GVCs, in which multinational firms fragment production and allocate different stages across countries based on cost structures and institutional conditions (Chinh et al., 2026). From this viewpoint, the economic effects of FDI are shaped not only by its volume but also by the way production activities are organized across borders. Evidence from Japanese investment in Southeast Asia also shows that FDI location decisions are closely related to host-market conditions and existing production networks. Kubo (2019) finds that expected consumption demand is particularly important for new Japanese FDI in Indonesia and Thailand, while the agglomeration of existing Japanese firms plays a stronger role in Malaysia and the Philippines. Accordingly, the role of the institutional environment becomes even more critical: improvements in governance can enhance regulatory predictability (Chinh et al., 2026; Nam et al., 2023), reduce uncertainty (Huang et al., 2026), and lower transaction costs (Xu et al., 2025), whereas political risk and institutional instability tend to deter foreign investment, especially in developing economies (Al Mustofa and Risyad, 2025). Furthermore, trade and investment agreements influence cross-border investment not only by facilitating market access but also through specific institutional provisions, such as investor protection clauses and dispute settlement mechanisms, which help strengthen policy credibility and mitigate uncertainty faced by multinational firms (Ossa et al., 2023).

2.1.3. Research gap

Despite this growing literature, two important gaps remain. First, most studies on foreign direct investment (FDI) and economic growth focus on observed capital inflows or the determinants of cross-border investment, rather than on whether investment-related

policy changes themselves generate measurable macroeconomic effects. Previous evidence further indicates that the growth effects of FDI may depend on the regulatory environment (Adams and Opoku, 2015), while the literature remains inconclusive as to whether international investment agreements consistently attract FDI inflows (Gopalan et al., 2023). Second, studies on deep regional agreements have mainly examined trade flows (Gavrila-Paven, 2026), FDI (Nhuong et al., 2025), multinational production (Miroudot and Rigo, 2022), and global value chain participation (Laget et al., 2020), while the aggregate growth effects associated specifically with their investment-related provisions have received less direct attention. Ma (2022) provides quasi-experimental evidence that broader economic integration under the Belt and Road Initiative can improve economic performance. Still, the growth consequences of investment-related institutional provisions within a specific regional agreement remain less clearly identified. From an institutional perspective, however, economic outcomes are closely linked to the quality, credibility, and predictability of policy frameworks (Siriopoulos et al., 2025). Investment-related provisions in regional agreements may therefore influence growth not only by affecting capital flows, but also by improving the investment environment, reducing uncertainty, and strengthening production-network coordination. This study addresses these gaps by treating the implementation of the ASEAN–Japan Comprehensive Economic Partnership Agreement (AJCEP) as a quasi-natural institutional shock and examining whether investment-related commitments embedded in a regional agreement generate measurable effects on economic growth.

2.2. From bilateral trade to a regional investment framework: the institutional background of AJCEP

The institutional evolution of regional integration in East Asia has been embedded in the expansion of cross-border production networks rather than driven solely by tariff reductions (Yu and Hu, 2026). Since the 1980s, Japanese manufacturing firms have continuously allocated different stages of production across East Asian economies, including parts production, processing, and assembly (Dzienis and McCaleb, 2025). A regional division of labor, supported by intermediate goods trade and cross-border investment, has gradually taken shape (Nguyen and Wu, 2021). ASEAN has become increasingly important within this system (Zhong and Su, 2021). On the one hand, its cost structure and capacity to absorb industries are suitable for the relocation of manufacturing activities. On the other hand, its expanding market size and geographical advantages have made it a key node in the regional strategies of Japanese firms. Studies on East Asian production networks generally show that cross-border investment, parts trade, and the deepening of regional division of labor have evolved simultaneously in this region (Huang, 2026). Firms' location choices are consistently shaped by the joint constraints of institutional environments, logistics efficiency, and market access conditions (Imai, 2020).

Before the signing of AJCEP, institutional linkages between Japan and ASEAN relied mainly on bilateral economic partnership agreements and relatively fragmented regional rule-of-law arrangements (Dent, 2005). Although these arrangements promoted trade and economic exchanges, they also led to problems of regulatory fragmentation. Differences across agreements in rules of origin, tariff preference eligibility, customs procedures, and regulatory requirements increased the institutional costs for firms when organizing production and allocating investment within the region. Rosenbaum (2025) refers to the "noodle bowl effect", which captures how overlapping agreements and complex rule application constrain firm behavior. For manufacturing firms that rely on cross-border division of labor, such institutional frictions extend beyond trade facilitation issues. They also directly affect the stability and efficiency of investment allocation.

The signing of AJCEP changed this institutional structure. According to official information from the Ministry of Foreign Affairs of Japan and ASEAN, AJCEP was signed on April 14th, 2008, and entered into force on December 1st, 2008 (AJCEP, 2008). Official statements clearly define it as a comprehensive framework for cooperation that promotes both trade and investment within the region. After its implementation, the entry into force among member states progressed in stages. Related institutional arrangements were further expanded to services and investment through subsequent protocol amendments. AJCEP is not merely an agreement on trade in goods. It represents a key institutional step in Japan's transition of its ASEAN economic cooperation from bilateral, fragmented arrangements to a more unified regional framework.

The significance of this institutional change does not lie in whether a specific investment provision was fully implemented at a single point in time. Instead, it lies in the substantial improvement in the predictability of regional cooperation, the degree of rule coordination, and policy continuity. For cross-border manufacturing investment, which involves high sunk costs and long-term allocation decisions, institutional stability itself is an important constraint on investment decisions. The impact of preferential trade agreements on FDI is not limited to market access in a narrow sense. It also operates by reducing policy uncertainty, lowering institutional transaction costs, and strengthening expectations of regional cooperation. AJCEP reflects such a restructuring of the regional investment environment. Firms allocating production and capital between Japan and ASEAN now face a more unified rule framework, rather than one primarily shaped by fragmented agreements.

2.3. Potential theoretical pathways for the growth effects of regional investment agreements

Investment policy does not directly generate output (Li et al., 2023). Its economic effects depend on whether institutional changes can alter firms' cross-border investment behavior and the production organization on which it relies. In the context of East Asia, this issue cannot be reduced to the general question of whether increased FDI inflows promote growth. Regional investment policies are embedded within production networks (Khan et al., 2024). Their effects are typically reflected in the joint adjustment of investment expectations, location choices, industrial chain organization, and domestic institutional responses (Chen et al., 2026). Compared with studies that treat FDI flows as the main shock variable, institutional investment policies function more as upstream constraints on firm behavior. The resulting growth effects mainly emerge through investment expansion, productivity improvement, and the differentiated release of institutional absorptive capacity (Farooq et al., 2026).

Changes in investment-related institutions may influence growth not only through capital accumulation but also by shaping expectations, reducing uncertainty, and improving the coordination of economic activities (Chinh et al., 2026; Huang et al., 2026; Xu et al., 2025). In this sense, investment policy affects both the scale and direction of capital flows while simultaneously strengthening regional production networks and enhancing firms' investment expectations. Examining the economic effects of investment policy changes from this institutional perspective is therefore essential for a comprehensive understanding of how regional economic cooperation shapes long-term development.

2.3.1. Credible institutional commitment and the investment expansion effect

Cross-border direct investment is characterized by long investment cycles, strong asset specificity, and high sunk costs (Cheng et al., 2026). It therefore relies more heavily on institutional stability and policy predictability than short-term capital flows. When there is substantial uncertainty about future rule enforcement, market access, or regulatory arrangements, firms may delay or scale back investment or relocate production to lower-institutional-risk locations (Chan and Yang, 2022). Institutional economics suggests that long-term transactions and capital investment depend on the stability and enforceability of rules (Haddad and Verriest, 2025).

At the international level, one key function of agreement arrangements is to enhance the credibility of policy commitments. Such arrangements help mitigate time-inconsistency in host-country policies and reduce investor concerns about potential institutional reversals. This effect is particularly evident in developing economies (Singh, 2026). J. Wang et al. (2026) show that international agreements can promote FDI inflows in developing economies by strengthening policy credibility and reducing institutional uncertainty. This effect arises not solely from legal protection clauses but also from the stable expectations generated by institutionalized cooperation.

Medvedev (2012) further finds that preferential trade agreements significantly increase FDI inflows. This effect is driven not only by tariff changes but also by improved expectations regarding market access and the policy environment. These findings indicate that the economic implications of such agreements extend beyond trade liberalization. They also influence how firms evaluate future investment returns and institutional risks.

From the perspective of growth theory, investment expansion remains a fundamental driver of macroeconomic growth (Bortolotti et al., 2025). In economies with relatively scarce capital and incomplete industrial upgrading, inflows of external capital can ease domestic savings constraints and expand fixed capital formation. It can also increase output through capacity expansion and reallocation of factors (Emako et al., 2023).

The underlying mechanism is not that policy directly drives growth. Instead, stronger institutional commitment improves firms' long-term investment incentives, thereby affecting growth through capital formation and investment expansion (Gamso and Grosse, 2021). The growth effect of regional investment policy first manifests in changes to the investment environment and subsequently in capital allocation and growth performance (Casi and Resmini, 2017).

2.3.2. Production network reorganization and productivity enhancement effects

The impact of investment policy is not limited to an increase in the quantity of capital. For regions such as RCEP that rely heavily on cross-border division of labor, the key issue is whether institutional changes alter the organization of production networks. The two-dimensional fragmentation framework proposed by Kimura and Ando (2005) suggests that firms distribute different stages of production across countries only when cross-border coordination costs are sufficiently low, the institutional environment is stable, and logistics and customs systems are predictable. Improvements in institutional coordination reduce the marginal cost of reorganizing supply chains and reallocating production activities within the region (Yu and Hu, 2026). This process facilitates the further deepening of production networks. Therefore, the growth effect of regional investment policy should not be understood solely in terms of FDI scale. It should be examined in the context of changes in the efficiency of production organization (Herold and Marzantowicz, 2023). When the institutional environment becomes more conducive to regional operations, firms tend to reorganize procurement, assembly, and export arrangements around existing production nodes. This process embeds more intermediate goods trade, parts processing, and upstream and downstream activities within regional value chains.

This mechanism is closely related to the literature on FDI spillovers. Lee (2025) notes that a key feature of East Asian trade is the rising share of production-network trade. This pattern indicates that trade and investment are highly complementary within the regional manufacturing system. The influence of institutional arrangements on cross-border investment also translates into changes in the regional division of labor. This logic is consistent with the literature on FDI spillovers. Nazzal et al. (2025) provide a comprehensive review and show that the economic effects of foreign investment mainly operate through demonstration effects, competition effects, labor mobility, and forward and backward linkages. Son and Kim (2025) further show that backward linkages are an important channel through which foreign investment improves the productivity of domestic firms. This evidence suggests that the growth effects of investment policy do not arise solely from increased capital. They also result from stronger firm linkages, finer division of labor, and higher productivity. When regional investment policy promotes the reorganization of production networks, it can generate more sustained growth effects through industrial chain integration, technology diffusion, and resource reallocation.

2.3.3. Conditional institutional gains: development stage and institutional environment

The economic effects of investment policy are neither automatic nor uniform. Whether external institutional arrangements translate into growth performance depends on the host country's ability to absorb and amplify the institutional gains generated by policy improvements (Arogundade et al., 2022). The literature on FDI and economic growth has shown that the growth effects of external capital are highly conditional. Guo et al. (2022) emphasize that human capital determines whether a host country can absorb the technological advantages associated with foreign investment. Appiah et al. (2023) show that the level of financial market

development affects whether foreign capital can be effectively transformed into economic growth. Qureshi et al. (2021) find that, after controlling for endogeneity and country heterogeneity, FDI does not necessarily lead to robust growth outcomes. The central implication is that the effects of external openness and foreign capital depend on domestic conditions rather than being universally constant.

The same logic applies to investment policy. When identical institutional arrangements are introduced across countries, their economic outcomes may differ. In middle- and low-income economies, where capital is relatively scarce, and there is greater scope for industrialization and structural transformation, the marginal returns from improvements in investment policy are often higher (Dinga et al., 2023). Once the regional investment environment improves, these economies are more likely to experience increases in capital formation, manufacturing expansion, and faster growth. In contrast, in economies with more abundant capital and more mature industrial structures, the marginal growth effects of similar policy shocks tend to be weaker (Siyakiya et al., 2023).

Differences in the institutional environment determine the efficiency of policy transmission. International agreements can provide external constraints, but their effectiveness depends on domestic institutional implementation (Buitrago R, 2025). Economies with stronger governance capacity, higher levels of market development, and more stable policy enforcement are better able to translate the cooperative signals generated by regional investment policies into observable and actionable expectations for firms (Garcia et al., 2024). By contrast, in economies with significant institutional frictions, unstable policy implementation, or weaker business environments, improvements at the regional level may not translate into actual investment responses. Therefore, the relationship between investment policy and economic growth is better understood as a form of conditional institutional gains (Zhou et al., 2023). The stage of development determines the potential magnitude of institutional benefits, while the institutional environment determines whether these benefits can be effectively realized (Thi Cam Ha et al., 2024).

2.4. Research hypotheses

By establishing high standards for investment and trade, AJCEP effectively reduces friction and uncertainty in cross-border economic activities. This logic is consistent with institutional economics, which emphasizes that the credibility and stability of formal rules shape investment incentives and long-term economic performance, and with recent evidence showing that preferential and investment-related agreements can influence FDI inflows, multinational production, and production-network organization (Gamso and Grosse, 2021; Miroudot and Rigo, 2022). These institutional gains promote inflows of foreign capital, facilitate the diffusion of advanced technology, and encourage improvements in the host country's investment environment (J. Wang et al., 2026). Together, these effects contribute to macroeconomic expansion. The three theoretical pathways outlined above provide the micro- and meso-level foundations for the expected effects of regional investment agreements. However, cross-country macroeconomic data do not allow the contribution of each channel to be isolated precisely because bilateral investment, sector-specific technology transfer, patent flows, and unobserved institutional changes are measured imperfectly. As a comprehensive institutional arrangement, AJCEP may affect growth through several interacting channels. Accordingly, the mechanisms discussed above are interpreted in this study as theoretically grounded channels that are consistent with the reduced-form effect of AJCEP. This interpretation is also consistent with recent evidence showing that deep regional agreements tend to jointly affect investment incentives, production network organization, and cross-border coordination.

Accordingly, this study adopts a reduced-form framework commonly used in macroeconomic policy evaluation and treats the implementation of AJCEP as a holistic institutional shock. The empirical analysis is designed to identify the agreement's overall effect on macroeconomic growth. Based on this framework, the study proposes the following hypotheses:

- **H1**: Holding other macroeconomic conditions constant, the implementation of the AJCEP, acting as a positive institutional shock, has significantly boosted macroeconomic growth in the signatory countries.
- **H2a**: The growth effects of investment policy shocks are expected to vary across countries at different levels of development.
- **H2b**: The growth effects of investment policy shocks are expected to vary across countries with different institutional and policy environments.

3. Research design and data construction

3.1. Sample and data sources

This study uses panel data from nine RCEP member economies over the period 2000–2023 to investigate the impact of the investment policy shock induced by the implementation of the ASEAN–Japan Comprehensive Economic Partnership Agreement (AJCEP) on economic growth. The selected countries include China, Korea, Australia, New Zealand, Vietnam, the Philippines, Malaysia, Thailand, and Indonesia. The chosen period provides sufficient pre- and post-policy observations to identify changes in economic performance resulting from the AJCEP's implementation.

The selection of AJCEP as the empirical case is appropriate for four reasons. First, AJCEP represents a clearly identifiable investment-related regional agreement between Japan and ASEAN economies, which allows its implementation to be treated as a policy shock in a quasi-experimental setting. Second, the agreement is closely connected to East Asian production networks, where Japan has long played a central role in coordinating trade, investment, and industrial specialization. Third, the timing of the 2008 implementation provides sufficient pre- and post-treatment observations over the 2000–2023 period, enabling evaluation of changes in macroeconomic performance before and after the policy intervention. Fourth, AJCEP directly affected ASEAN participants while leaving other RCEP economies outside the Japan–ASEAN institutional framework, thereby providing a meaningful basis for

constructing treatment and control groups.

Macroeconomic variables, such as GDP, investment, trade, and population, are primarily sourced from the World Bank's World Development Indicators (WDI) database, which provides long-term, internationally comparable, and high-quality data. Institutional indicators, capturing governance quality, regulatory effectiveness, and policy stability, are primarily sourced from the World Bank's Worldwide Governance Indicators (WGI), a widely recognized source of cross-country institutional measures in international economics research.

Before analysis, the data were systematically cleaned and harmonized to ensure reliability and comparability. This included cross-checking values across different sources, addressing inconsistencies, and supplementing a small number of missing observations using linear interpolation. Monetary values were converted into constant 2010 US dollars to maintain comparability over time. These steps ensure that the panel dataset is balanced and suitable for econometric analysis, particularly for applying a DID framework.

To further strengthen identification, an event-study analysis is conducted to assess the parallel-trends assumption. This step examines whether the treatment and control groups exhibited systematic differences in economic growth before the AJCEP implementation, thereby assessing the suitability of the selected control group as a counterfactual benchmark.

3.2. Treatment and control group selection

The identification of treatment and control groups is critical to the quasi-natural experimental design of this study. The treatment group consists of five ASEAN countries: Vietnam, the Philippines, Malaysia, Thailand, and Indonesia. These countries are direct participants in the Japan–ASEAN economic cooperation framework and are therefore directly affected by the investment-related institutional arrangements introduced under AJCEP. The agreement primarily aims to strengthen investment protection, trade facilitation, and regulatory coordination between Japan and ASEAN member states, and its 2008 implementation constitutes an exogenous policy shock expected to improve the investment environment and stimulate economic growth in these countries.

Certain ASEAN members are excluded from the treatment group to maintain internal validity and comparability. Singapore and Brunei are high-income economies with economic structures dominated by finance, services, and logistics, which differ markedly from those of manufacturing-oriented Southeast Asian economies. Including these countries could introduce structural heterogeneity and bias estimation results. Similarly, Cambodia, Laos, and Myanmar are excluded due to serious data limitations, as key macroeconomic indicators are not consistently available over the sample period. Restricting the treatment group to countries with comparable economic structures and reliable data ensures robust and internally consistent estimation of AJCEP's effects.

The control group comprises China, Korea, Australia, and New Zealand. Japan is not included in the control group because it is the agreement partner of AJCEP and therefore cannot serve as an untreated counterfactual for evaluating the Japan-ASEAN policy shock. While these countries are also RCEP members, they are not part of the Japan–ASEAN institutional framework and thus are not directly influenced by AJCEP. They provide a valid counterfactual for assessing the policy shock's causal impact on the treatment group. The inclusion of Australia and New Zealand, despite their high-income status, is justified by their strong economic integration with East and Southeast Asia through trade, investment, and synchronized macroeconomic cycles. Their industrial structures, while more advanced than those of the treatment group, still maintain significant manufacturing and trade linkages with the region, allowing them to serve as comparable benchmarks for capturing regional macroeconomic fluctuations.

The selection of control countries follows two guiding principles. First, control countries should operate within a similar regional economic environment to minimize confounding effects from global macroeconomic shocks. Second, control countries should not be directly affected by AJCEP's institutional arrangements, preserving their validity as a counterfactual. By carefully defining treatment and control groups based on institutional exposure, economic structure, and regional integration, this study establishes a solid foundation for causal inference using the DID approach.

3.3. Variable definitions

The selection of variables follows the theoretical focus of this study and the requirements of the DID identification strategy. Economic growth is used as the dependent variable because the study examines whether an investment-related institutional shock generates measurable macroeconomic effects. The AJCEP policy shock is captured by the DID interaction term, which identifies the difference in economic growth between treated and control economies following the agreement's implementation. The control variables are selected to account for major macroeconomic and structural factors that may simultaneously affect economic growth and correlate with policy exposure. Inflation captures macroeconomic stability; government effectiveness reflects the credibility and implementation capacity of public institutions; trade openness controls for external economic integration; and urbanization captures structural and demographic differences across countries. These variables help reduce omitted-variable bias while avoiding over-control for potential transmission channels such as FDI inflows or production-network adjustment.

3.3.1. Dependent variable

This study uses economic growth as the dependent variable and measures it by the annual GDP growth rate (%). The GDP growth rate reflects changes in the speed of economic expansion within a country over a given period. It is therefore suitable for analyzing the dynamic effects of institutional policy shocks on macroeconomic performance (Sungurtekin Hallam, 2022).

3.3.2. Key explanatory variable

The key explanatory variable in this study is the policy shock variable, denoted DID, which is used to identify the impact of AJCEP

implementation on economic growth in the relevant countries. Specifically, three dummy variables are constructed. $Treat_i$: is the treatment group indicator. It takes the value of 1 if a country belongs to the treatment group, and 0 otherwise. $Post_t$: is the post-policy dummy variable, which equals 1 for years $t \geq 2008$ and 0 otherwise. DID_{it}=$Treat_i \times Post_t$ is the interaction term. It captures the average treatment effect after AJCEP implementation. This variable reflects the difference in economic growth between the treatment and control groups after the implementation of AJCEP.

3.3.3. Control variables

To mitigate potential omitted-variable bias, the baseline model includes four control variables. The inflation rate (IRC), measured as the annual growth of the consumer price index, captures macroeconomic price fluctuations (Pham and Sala, 2022). Institutional quality (GOV) captures government effectiveness and the overall institutional environment, reflecting the capacity and credibility of policy implementation (Mehmood et al., 2023). Trade openness (OPEN), defined as the ratio of total imports and exports to GDP, captures the degree of integration into the global economy (Yu and Gu, 2025). Finally, the share of the urban population (URBAN) serves as a proxy for urbanization, indicating population structure and the spatial concentration of economic activity (Li et al., 2025). These controls are included because cross-country variations in economic growth are commonly influenced by macroeconomic stability, institutional quality, external openness, and structural characteristics, all of which could otherwise confound the estimated effect of investment policy. Table 1 provides a detailed explanation of these variables.

3.4. Model specification

To identify the impact of the AJCEP policy shock on economic growth, this study employs a DID model and includes both country- and year-fixed effects. The baseline model is specified as follows:

$$EP_{i,t} = \alpha_0 + \alpha_1(Treat_i \times Post_t) + \alpha_2 X_{i,t} + \eta_i + \varphi_t + \varepsilon_{i,t} \quad (1)$$

EP_{it} denotes the GDP growth rate of country i in year t. The term $Treat_i \times Post_t$ represents the policy interaction term. $X_{i,t}$ is a vector of control variables. α_0 is the constant term, and α_2 is the coefficient vector associated with the control variables. η_i denotes country fixed effects, φ_t denotes year fixed effects, and ε_{it} is the random error term.

The coefficient α_1 is the key parameter of interest in this study. It captures the change in economic growth in the treatment group relative to the control group after AJCEP implementation. Country fixed effects control for time-invariant country characteristics, such as geographical conditions, institutional traditions, and industrial structure. Year fixed effects control for common macroeconomic shocks that affect all countries, such as changes in the global economic cycle.

3.5. Identification strategy

The validity of the DID approach depends critically on the parallel trends assumption, which underlies causal identification in this framework. Specifically, the DID method attributes treatment effects to differential changes in outcomes between treated and control units, assuming that, in the absence of the intervention, both groups would have followed similar economic growth trajectories. To assess this assumption, the study employs an event-study design to estimate dynamic effects before and after policy implementation. Lead and lag terms relative to the timing of the policy are constructed to examine whether there were systematic differences between the treatment and control groups before the intervention. When the parallel trends assumption is satisfied, observed post-policy differences in economic growth can be credibly interpreted as the effect of AJCEP.

The DID framework is selected because it is well-suited to evaluating policy interventions that affect a clearly defined treatment group while leaving a comparable control group untreated (Angrist and Pischke, 2009; Callaway and Sant'Anna, 2021). Compared with simple before-and-after comparisons, DID removes common time shocks that affect all countries. Compared with cross-sectional regressions, it controls for time-invariant country characteristics through country fixed effects. It is also more appropriate than a purely synthetic-control design in this setting because the study involves multiple treated economies and a limited donor pool. Nevertheless,

Table 1
Variable Definition.

Variable Symbol	Variable	Variable Definition
EP	Economic Growth	Uses GDP growth rate (%) to measure the rate of economic expansion, reflecting the overall dynamics and trends of economic activities.
DID2008	AJCEP Policy Shock	Interaction term equal to $Treat_i \times Post_t$, where $Treat_i = 1$ for treatment-group countries and 0 otherwise, and $Post_t = 1$ for years 2008 and after and 0 otherwise. This variable captures the average policy effect of AJCEP implementation on economic growth in the treated countries relative to the control group.
IRC	Inflation Rate	Annual percentage change in the Consumer Price Index (CPI), measuring yearly consumer-price inflation.
GOV	Institutional Quality	Measured by the Government Effectiveness percentile rank from the Worldwide Governance Indicators (WGI), ranging from 0 to 100. This indicator reflects the quality of public services, policy formulation and implementation, and the credibility of government policy commitments.
OPEN	Trade Openness	Proportion of total trade volume to GDP, reflecting the degree of economic openness.
URBAN	Urbanization Level	Share of urban population in total population (%), indicating the level of urbanization.

DID has important limitations. Its validity depends on the parallel-trends assumption, and the estimates may be affected by concurrent regional agreements, global shocks, or differences in treatment timing (Goodman-Bacon, 2021). To address these concerns, this study uses an event-study specification to examine pre-treatment trends, includes country and year fixed effects, controls for competing regional agreements in robustness checks, tests alternative policy timing, excludes major crisis years, applies PSM-DID to improve covariate comparability, and conducts placebo tests to assess whether the estimated effect is driven by random assignment or spurious time trends. The detailed results of these diagnostic and robustness procedures are reported in the subsequent empirical sections.

4. Empirical results

4.1. Descriptive statistics

The descriptive statistics of the main variables are presented in Table 2. After sample selection, the final dataset includes nine countries, namely China, Korea, Australia, New Zealand, Vietnam, the Philippines, Malaysia, Thailand, and Indonesia, covering the period from 2000 to 2023. Although the country dimension is limited, the balanced country-year panel provides sufficient temporal variation for examining changes before and after AJCEP implementation within the selected regional sample. For the dependent variable, economic growth (measured by the GDP growth rate), the mean is 4.64%, the minimum is −9.52%, the maximum is 14.15%, and the standard deviation is 2.87%. These results indicate substantial fluctuations in economic growth across countries during the sample period. Such variation reflects differences in developmental stages and macroeconomic conditions across countries and suggests sufficient variation along both the time and country dimensions.

For the control variables, the inflation rate measured by the consumer price index has a mean of 3.29%, with values ranging from negative 1.61% to 23.12%. This indicates moderate inflation, with substantial variation across observations. The average value of government effectiveness (GOV) is 69.77%, with substantial variation in institutional quality across countries. Trade openness (OPEN) has a mean of 85.97% and a standard deviation of 45.55%, suggesting that countries are generally highly open, although the degree of openness varies considerably. The average level of urbanization (URBAN) is 60.83%, indicating that the sample countries exhibit moderate-to-high urbanization levels, with substantial cross-country variation. Overall, the variables exhibit substantial cross-country variation, providing a solid data foundation for the subsequent empirical analysis.

4.2. Multicollinearity test

To assess potential multicollinearity among the explanatory variables, this study employs the variance inflation factor (VIF) for diagnostic analysis (Table 3). The VIF measures the extent to which linear relationships among the explanatory variables inflate the variance of an estimated coefficient. It is defined as $VIF = 1 / (1-R^2)$, where R^2 is the coefficient of determination obtained by regressing the variable on all other variables. A VIF value greater than 10 is generally considered to indicate a serious multicollinearity problem.

The results show that the VIF values for all variables are below the conventional threshold of 10. The minimum value is 1.22, the maximum value is 8.58, and the sample mean is 4.16. These findings indicate that there is no significant multicollinearity problem in the model. The results do not indicate severe multicollinearity under the conventional threshold. Specifically, the VIF values for urbanization (URBAN) and government effectiveness (GOV) are 8.22 and 8.58, respectively. Although these values are relatively high, they remain within an acceptable range. This pattern may reflect an association between the institutional environment and population structure in the sample countries. Countries with stronger governance capacity tend to exhibit higher levels of urbanization (Ye et al., 2024). In contrast, the VIF values for the remaining variables are relatively low. The VIF values for inflation (IRC) and trade openness (OPEN) are 1.22 and 1.43, respectively, and the VIF value for the policy variable (DID2008) is 1.33. These results indicate that there is no strong linear dependence among these variables. Therefore, it can be concluded that the model does not suffer from serious multicollinearity, and the correlations among explanatory variables do not materially affect the stability of the regression estimates.

4.3. Baseline DID estimates

This study treats the 2008 signing and implementation of the ASEAN–Japan Comprehensive Economic Partnership Agreement as a quasi-natural institutional shock. It applies a DID approach to evaluate the impact of investment policy shocks on economic growth. Table 4 reports the baseline regression results. Model (1) includes only the policy variable DID. Model (2) adds control variables.

Focusing on the key explanatory variable, the coefficient of the policy interaction term DID2008 is positive and statistically

Table 2
Descriptive Statistics.

	N	Mean	SD	Min	Max
EP	216	4.641	2.865	−9.518	14.15
IRC	216	3.288	2.892	−1.611	23.115
GOV	216	69.774	17.667	32.432	98.571
OPEN	216	85.973	45.548	32.972	220.407
URBAN	216	60.833	19.869	24.374	86.985

Table 3
Variance Inflation Factors.

VARIABLES	VIF	1/VIF
URBAN	8.224	0.122
GOV	8.583	0.117
OPEN	1.427	0.701
IRC	1.220	0.819
DID2008	1.330	0.752
Mean VIF	4.157	

Table 4
Baseline Regression Results.

	(1) EP	(2) EP
DID2008	0.995**	1.630***
	(2.039)	(3.199)
IRC		0.034
		(0.557)
GOV		0.028
		(0.996)
OPEN		0.004
		(0.462)
URBAN		−0.160***
		(−4.005)
Constant	5.983***	12.456***
	(10.687)	(4.749)
Observations	216	216
Number of Country ID	9	9
Adjusted R-squared	0.470	0.506
Country FE	YES	YES
Year FE	YES	YES

Note: t-statistics are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

significant in both specifications. Without control variables, the coefficient is 0.995 and is significant at the 5 percent level. After including control variables, the coefficient increases to 1.630 and becomes significant at the 1 percent level. These results indicate that, after the implementation of AJCEP, economic growth in the treatment group increased significantly relative to the control group. This finding suggests that regional investment institutional arrangements, as represented by AJCEP, may support economic growth by improving investment conditions and encouraging productive cross-border investment. Uttama (2021) finds that investment provisions in international investment agreements are associated with higher inward FDI in RCEP economies. In the preferred specification, the coefficient of 1.630 implies that annual GDP growth in the treated economies increased by approximately 1.63 percentage points relative to the control group following the implementation of AJCEP. Greater predictability of investment rules can reduce the policy uncertainty surrounding long-term cross-border investment and strengthen firms' incentives to commit capital. Using data on Japanese multinational enterprises, Inada and Jinji (2024) show that investment-related policy uncertainty reduces FDI. The extent to which additional investment translates into economic growth also depends on the domestic institutional environment. Adams and Opoku (2015) find that effective regulation strengthens FDI's contribution to economic growth. In this context, AJCEP can support growth by strengthening firms' incentives to undertake cross-border investment and by providing a more predictable institutional environment in which such investment can contribute to productive economic activity. From an identification perspective, the estimates are obtained after controlling for country- and year-fixed effects. This approach removes the influence of time-invariant country characteristics and common macroeconomic shocks, which allows for a more reliable identification of the average treatment effect of the policy shock.

A comparison between columns (1) and (2) shows that, after the inclusion of control variables, the coefficient of DID2008 remains significant and increases in magnitude. This result indicates that the baseline finding is less likely to be driven solely by the included observable covariates. After controlling for inflation, institutional quality, trade openness, and urbanization, the positive effect of investment policy shocks on economic growth remains robust. Specifically, the DID coefficient increases from 0.995 to 1.630 after these controls are introduced. Its positive sign and statistical significance are preserved after accounting for differences in macroeconomic stability, institutional conditions, external openness, and structural development. The estimated post-AJCEP growth differential, therefore, remains evident after these observable factors are taken into account.

Regarding the control variables, government effectiveness (GOV), inflation (IRC), and trade openness (OPEN) all have positive but statistically insignificant coefficients. This suggests that their independent effects on economic growth are not clearly identified in the baseline specification after controlling for country and year fixed effects and other macroeconomic factors. For IRC, the result indicates that short-run changes in inflation do not independently explain the growth differential identified in the baseline model. The insignificant GOV coefficient is also compatible with the institutional mechanism developed in this study. Government effectiveness enters

the regression as an aggregate control, while the relevant mechanism concerns how institutional conditions influence the transmission of investment into productive activity. Adams and Opoku (2015) report a closely related result: FDI and regulation are individually insignificant in their growth regressions, whereas their interaction has a significant positive effect, indicating that the regulatory environment conditions the growth contribution of foreign investment. For OPEN, the insignificant coefficient indicates that aggregate trade openness alone does not explain the estimated growth effect. A high trade-to-GDP ratio does not necessarily capture how investment and production are organized across borders. The growth implications of regional integration depend more on whether foreign investment is embedded in production networks and value-chain linkages that support specialization, technology transfer, and productivity gains. Son and Kim (2025) show that the productivity effects of FDI vary with firms' position in global value chains and the technological composition of foreign investment, highlighting the importance of how investment is integrated into cross-border production structures.

The coefficient of urbanization (URBAN) is negative and significant at the 1% level. This result indicates that increases in urbanization are associated with lower annual GDP growth after controlling for the other macroeconomic factors and fixed effects. One possible explanation is that the growth effects of urbanization are not uniform across economies and depend on how efficiently urban expansion translates into productivity gains and structural transformation. Shaban et al. (2024) show that the relationship between urbanization and economic growth varies across countries and may involve reverse causality, suggesting that higher urbanization alone does not necessarily lead to stronger contemporaneous growth. In the present fixed-effects model, persistent cross-country differences in urbanization are absorbed by country effects, so the estimated coefficient mainly reflects within-country changes over time. Although URBAN may be related to broader macro-level structural factors, including government effectiveness and resource allocation patterns, the VIF results indicate that the model does not suffer from serious multicollinearity. The estimated effect of the core policy variable DID2008 remains positive and statistically significant after the inclusion of these controls.

The baseline model includes both country fixed effects and year fixed effects. Given the panel structure of the data and the possibility of within-country serial correlation, an additional inference check is conducted for the preferred baseline specification reported in Column (2). As shown in Appendix A1, the model is re-estimated using country-clustered standard errors, and the DID2008 coefficient remains positive and statistically significant, with a country-clustered t-statistic of 3.999. Wild cluster bootstrap inference is also applied with 9999 replications and country-level clustering. The resulting bootstrap p-value is 0.0173, and the 95% confidence interval is [0.5678, 2.7190]. These results indicate that the main DID finding is not driven by conventional fixed-effects inference and remains robust after accounting for within-country serial correlation and small-cluster inference.

Overall, the baseline regression results show a significant positive relationship between the AJCEP policy shock and economic growth. This finding supports H1, indicating that the implementation of AJCEP, acting as a positive institutional shock, has significantly boosted macroeconomic growth in the treated economies. The estimated effect is economically meaningful and remains robust after controlling for key macroeconomic and structural factors. The results suggest that investment-related regional cooperation can generate measurable growth gains when institutional improvements strengthen firms' willingness to invest and support the productive use of cross-border capital.

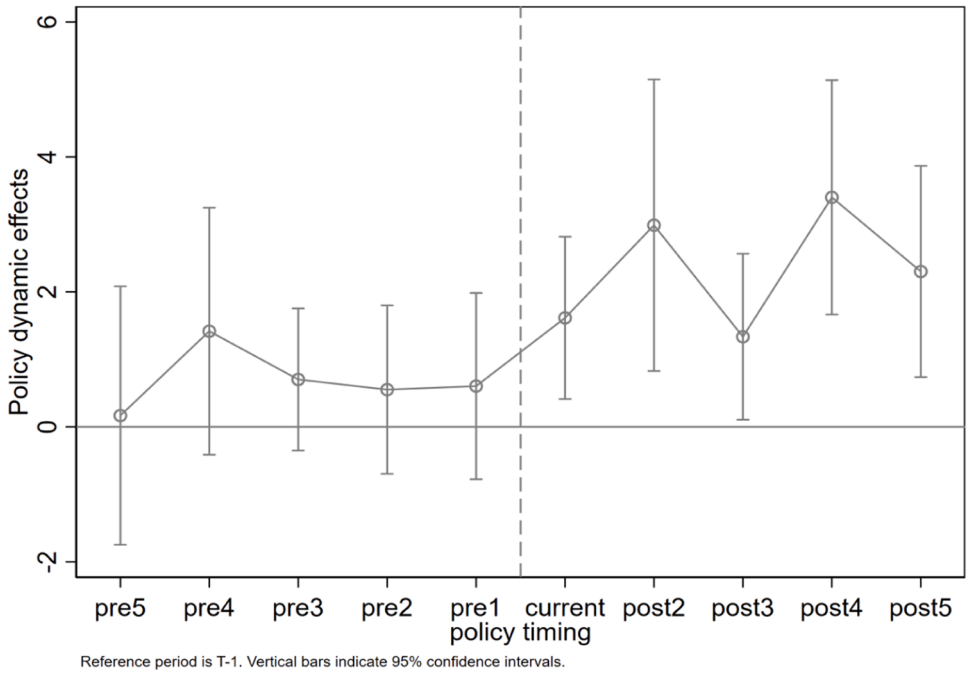


Fig. 1. Parallel Trend Graph.

4.4. Parallel trend test

The validity of the DID approach depends on the parallel trends assumption. This assumption implies that before AJCEP implementation, there were no systematic differences in economic development trends between the treatment and control groups. The growth rates of the two groups should follow similar trajectories over time. To test this identification condition, the study follows the event-study approach commonly used in the literature to capture dynamic policy effects (Jacobson et al., 1993). Time dummy variables relative to the policy implementation are constructed to estimate the dynamic effects of AJCEP before and after its implementation. By examining whether the coefficients in the pre-policy periods significantly differ from zero, the study tests whether the sample satisfies the parallel-trends assumption and further analyzes the dynamic impact of investment policy shocks on economic growth.

Fig. 1 illustrates the dynamic effects of AJCEP on economic growth. The base period is set to 2008, the year of policy implementation. The variables pre5, pre4, pre3, pre2, and pre1 represent the treatment group dummy variables for five, four, three, two, and one years before the policy, respectively. The variables post2, post3, post4, and post5 represent the second to fifth years after the policy implementation. The confidence intervals for all five pre-policy coefficients include zero, indicating that the pre-policy estimates are not statistically different from zero. This result suggests that there are no significant differences in economic growth trends between the treatment and control groups before policy implementation, supporting the parallel-trends assumption.

Fig. 1 reports the dynamic effects around AJCEP implementation. The coefficients before policy implementation are not statistically different from zero, and the joint test of the pre-treatment coefficients is insignificant, with $p = 0.2759$. This indicates that there is no evidence of systematic differences in economic growth trends between the treatment and control groups before AJCEP implementation, supporting the parallel-trends assumption. After policy implementation, the estimated coefficients generally increase, suggesting that the growth effect becomes more visible once the policy takes effect. The gradual emergence of the post-treatment effect is economically plausible in the context of an investment-related regional agreement. Changes in investment rules can alter firms' expectations relatively quickly, whereas cross-border capital commitments and the resulting expansion of productive activity require time to materialize. Chong and Srebot (2020) report a similar dynamic pattern for regional trade agreements, finding that their effect on investment confidence is not immediate but develops over time. The post-AJCEP pattern, therefore, suggests that the growth response strengthened as firms had time to adjust investment decisions to the new regional institutional environment. In addition, Appendix A2 reports an event-study check based on country-specific AJCEP implementation years. The pre-treatment coefficients are also jointly insignificant ($p = 0.2295$), further supporting the validity of the identification strategy under the staggered implementation setting.

4.5. Robustness checks

To further examine the robustness of the baseline regression results, this study conducts additional tests by excluding the effects of the pandemic, controlling for competing regional policy shocks, allowing for country-specific AJCEP implementation timing, and

Table 5
Robustness Regression Results.

VARIABLES	(1) EP	(2) EP	(3) EP	(4) EP	(5) EP	(6) EP	(7) EP
DID2008	2.064***	1.312*	2.838***	2.573***	1.245***	1.877***	1.873***
	(4.620)	(1.955)	(3.824)	(2.909)	(2.644)	(3.637)	(3.587)
Treat x GFC							−1.560*
							(−1.879)
IRC	−0.002	0.031	0.030	0.028	0.015	0.067	0.050
	(−0.031)	(0.519)	(0.501)	(0.473)	(0.246)	(0.938)	(0.830)
GOV	0.015	0.026	0.032	0.030	0.025	0.034	0.031
	(0.553)	(0.892)	(1.146)	(1.059)	(0.860)	(1.167)	(1.092)
OPEN	0.002	0.005	0.007	0.008	0.002	0.005	0.004
	(0.310)	(0.558)	(0.836)	(0.896)	(0.217)	(0.667)	(0.538)
URBAN	−0.159***	−0.182***	−0.176***	−0.192***	−0.155***	−0.158***	−0.168***
	(−4.386)	(−3.623)	(−4.374)	(−3.840)	(−3.836)	(−4.024)	(−4.199)
Constant	13.500***	13.776***	12.773***	13.764***	12.629***	11.768***	12.588***
	(5.471)	(4.320)	(4.915)	(4.360)	(4.763)	(4.482)	(4.831)
AKFTA policy	NO	NO	YES	YES	NO	NO	NO
ACFTA policy	NO	YES	NO	YES	NO	NO	NO
Policy timing	Uniform 2008	Uniform 2008	Uniform 2008	Uniform 2008	Country-specific	Uniform 2008	Uniform 2008
Observations	198	216	216	216	216	198	216
Number of countries	9	9	9	9	9	9	9
Adjusted R-squared	0.322	0.504	0.516	0.514	0.497	0.524	0.512
Country FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES

Notes: t-statistics are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All specifications include country and year fixed effects. Column (5) uses country-specific AJCEP implementation timing, although the coefficient is reported under DID2008 for consistency.

accounting for the potential influence of the global financial crisis.

Column (1) of Table 5 reports the regression results after excluding the years 2020 and 2021, which were affected by the pandemic. The results show that the coefficient of the policy interaction term DID2008 remains positive and statistically significant at the 1 percent level. This finding indicates that after accounting for the adverse external shock of the COVID-19 pandemic, the positive growth effect associated with the AJCEP implementation remains statistically significant.

Columns (2), (3), and (4) of Table 5 present the regression results after including dummy variables for competing policy shocks, namely the ASEAN–Korea Free Trade Agreement in 2007 (The ASEAN Secretariat, 2007) and the ASEAN–China Free Trade Agreement in 2010 (The ASEAN Secretariat, 2010), both separately and jointly. The results show that, after controlling for the effects of these competing policies, the coefficient of the policy interaction term DID2008 remains positive and statistically significant at conventional levels. This result indicates that the estimated AJCEP effect persists after accounting for ACFTA and AKFTA. These agreements were implemented in a similar period and could also have affected economic growth in ASEAN economies. Their inclusion, therefore, helps separate the AJCEP effect from other contemporaneous regional integration shocks. The positive and significant DID2008 coefficient reduces the concern that the baseline result mainly captures the effects of ACFTA or AKFTA.

To address the concern that AJCEP entered into force at different dates across member countries, Column (5) reconstructs the DID variable using country-specific implementation years. Specifically, Vietnam is coded as treated from 2008, Malaysia and Thailand from 2009, and the Philippines and Indonesia from 2010. The coefficient remains positive and statistically significant at the 1 percent level, suggesting that the uniform 2008 policy-year coding does not drive the baseline result. In addition, Appendix A2 reports the country-specific event-study results, where the pre-treatment coefficients are jointly insignificant ($p = 0.2295$), further supporting the validity of the identification strategy under the staggered implementation setting.

Columns (6) and (7) further examine whether the estimated AJCEP effect is affected by the overlap between AJCEP implementation and the 2008 global financial crisis. Column (6) excludes the crisis years 2008 and 2009 from the estimation sample. The DID coefficient remains positive and statistically significant at the 1 percent level, with a coefficient of 1.877 and a t-statistic of 3.637. This indicates that observations from the immediate crisis period do not drive the estimated policy effect. Column (7) includes an interaction between the treatment-group indicator and the global financial crisis dummy to capture treatment-specific crisis-period shocks. The DID coefficient remains positive and statistically significant at the 1 percent level, with a coefficient of 1.873 and a t-statistic of 3.587. The treatment-specific crisis interaction is negative and weakly significant at the 10 percent level, suggesting that the treated economies did not experience a more favorable crisis-period shock relative to the control group. Instead, the negative interaction indicates relatively weaker growth in the treated economies during the crisis period, making a favorable crisis shock an unlikely explanation for the positive post-AJCEP growth differential. These results indicate that the positive AJCEP effect remains robust after accounting for the potential influence of the global financial crisis.

4.6. PSM-DID

Because AJCEP exposure is not randomly assigned, systematic differences in observable characteristics between the treatment and control groups may affect the estimated policy effect. The baseline model controls for country-level characteristics and includes country and year fixed effects. These fixed effects account for time-invariant country heterogeneity and common macroeconomic shocks, but they do not directly balance the distribution of observable covariates across the two groups. PSM-DID is therefore used as an additional robustness check to improve covariate comparability before re-estimating the policy effect.

To address this issue, following the PSM-DID approach used in policy evaluation studies (Duong and Thanh, 2019), this study combines propensity score matching with the DID approach to construct a PSM-DID model for robustness analysis (Table 6). First, the control variables used in the baseline regression are treated as covariates. A logit model is applied to estimate propensity scores. Based on these scores, a nearest-neighbor one-to-one matching procedure is used to construct comparable treatment and control groups.

On this basis, the impact of AJCEP on economic growth is re-estimated, and the results are reported in Table 7. The results show that, both before and after the inclusion of control variables, the policy variable DID remains positive and statistically significant at the 5 percent and 1 percent levels. This finding indicates that the estimates obtained from the PSM DID approach are consistent with the baseline regression results. The preferred PSM-DID coefficient is 1.764, close to the baseline estimate of 1.630, even after matching

Table 6
Covariate Balance Test after Propensity Score Matching.

Variable	Unmatched/matched	Mean		% reduct |bias|	*t*-test	
		Treated	Control		t	p>|t|
IRC	U	3.121	2.904	48.5	0.57	0.573
	M	3.135	3.247		−0.24	0.809
GOV	U	65.918	75.857	88.0	−4.36	0.000
	M	66.553	65.361		0.47	0.638
OPEN	U	93.018	72.125	96.1	3.50	0.001
	M	92.229	93.036		−0.11	0.913
URBAN	U	57.976	67.385	79.3	−3.54	0.000
	M	58.242	56.29		0.65	0.516

Table 7
PSM-DID Regression Results.

	(1) EP	(2) EP
DID2008	1.301**	1.764***
	(2.305)	(3.016)
IRC		0.042
		(0.622)
GOV		0.034
		(1.026)
OPEN		0.002
		(0.211)
URBAN		−0.157***
		(−3.588)
Constant	6.251***	11.994***
	(10.060)	(4.214)
Observations	190	190
Number of Country ID	9	9
Adjusted R-squared	0.479	0.509
Country FE	YES	YES
Year FE	YES	YES

Note: t-statistics are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

substantially reduces the initial differences in government effectiveness, trade openness, and urbanization. This weakens the explanation that the positive post-AJCEP growth differential is mainly driven by pre-existing observable differences between the two groups. Li et al. (2024) find that high-quality investment treaties significantly increase multinational firms' propensity to invest in a host country, and that this effect strengthens after the treaties enter into force. Their firm-level evidence shows that the quality and effective implementation of investment agreements can materially alter firms' investment decisions, providing a plausible channel through which AJCEP-related institutional changes can affect economic activity.

The PSM-DID results mitigate observable differences between the treatment and control groups, while the estimated effect may still be affected by how the treatment and control samples are defined. To further examine this issue, this study conducts two sample-composition robustness checks to assess whether the main result is sensitive to plausible sample adjustments. First, Australia and New Zealand are excluded from the control group. These two economies are relatively high-income, structurally more mature control economies, and may differ from the treated ASEAN economies in terms of their development stage and economic structure. This check, therefore, examines whether the baseline finding is driven by the inclusion of these two high-income control economies. The results show that, after excluding Australia and New Zealand, the DID2008 coefficient remains positive and statistically significant at the 1 percent level, with a coefficient of 2.325 and a t-statistic of 3.588. This indicates that the positive growth effect of AJCEP does not rely on the inclusion of Australia and New Zealand as control economies.

Second, Singapore and Brunei are included in an expanded treatment group to examine whether excluding these two ASEAN economies affects the main finding. Singapore is a highly open service- and logistics-oriented economy, while Brunei is a small, high-income, resource-based economy. Their exclusion from the baseline treatment group is intended to preserve comparability in the development stage and production structure among the treated economies. The expanded-treatment-group check shows that the DID2008 coefficient remains positive and statistically significant at the 5 percent level, with a coefficient of 1.140 and a t-statistic of 2.091. This suggests that the positive growth effect of AJCEP remains observable when Singapore and Brunei are included in the treated group. The coefficient is smaller than the baseline estimate of 1.630. This attenuation suggests that the average growth response is weaker when economies with markedly different production structures are added to the treatment group. In particular, service-intensive and resource-based economies may translate regional investment commitments into domestic production and growth differently from the more manufacturing-oriented economies in the baseline treatment group. The result, therefore, points to the domestic production structure as one factor shaping the magnitude of the AJCEP growth effect. These results indicate that the main DID finding remains robust after adjusting for differences in the composition of both the control and treatment groups. The detailed estimates are reported in Appendix A3 and Appendix A4.

4.7. Placebo test

To examine whether the implementation of AJCEP generates spurious effects driven by random treatment assignment or time variation, this study conducts a placebo test by randomly assigning a pseudo-treatment group within the estimation sample. Since Japan is excluded from the main estimation sample, the placebo test is conducted among the remaining nine economies. In each replication, five economies are randomly selected as the pseudo-treatment group, and the remaining four economies are assigned to the pseudo-control group, thereby preserving the treatment-group size used in the baseline DID specification. The pseudo-treatment indicator is then interacted with the post-2008 time dummy to construct a placebo DID variable, and the baseline fixed-effects model is

re-estimated. This random assignment process is repeated 1000 times. The random draws are conducted with replacement so that the same pseudo-treatment allocation may appear more than once. The placebo coefficients are concentrated around zero and clearly separated from the baseline estimate, indicating that the baseline result is unlikely to be driven by random treatment assignment.

Fig. 2 presents the distribution of the estimated coefficients with economic growth as the dependent variable. The results show that the estimated coefficients are concentrated around zero and follow an approximately normal distribution. Most p-values are greater than 0.1, indicating that the randomly generated treatment assignments do not have significant effects on economic growth. Therefore, the placebo evidence supports the robustness of the baseline DID estimates. This finding suggests that random factors do not drive the identified policy effects and that the effects are statistically reliable.

4.8. Heterogeneity analysis

Foreign direct investment is not a universal driver of economic growth under all conditions. Its effects are strongly conditioned by the host country's economic characteristics and institutional environment (Khan et al., 2024). Therefore, this study conducts heterogeneity analysis along two dimensions, income level and industrial-policy orientation, to examine whether the growth effects of investment policy shocks differ across development stages and policy-orientation environments.

The heterogeneity analysis is conducted along two dimensions: income level and industrial-policy orientation. The income classification is based on the relative pre-AJCEP income position of the sample economies, using GDP per capita information from the World Development Indicators (WDI). Australia, New Zealand, Korea, China, Malaysia, and Thailand are classified as the higher-income group (HI), while Vietnam, the Philippines, and Indonesia are classified as the middle- and lower-income group (MLI).

The industrial-policy orientation classification distinguishes between interventionist industrial-policy economies (IIP) and liberal market-oriented economies (LME). This distinction is informed by the comparative-capitalism literature on liberal market economies and by the industrial-policy literature on state coordination, selective policy intervention, and industrial upgrading (Evans and Evans, 2012; Hall and Soskice, 2001; Rodrik, 2008; Wade, 1990). China, Korea, Malaysia, Thailand, and Vietnam are classified as IIP economies because their pre-AJCEP development strategies relied more strongly on state guidance, industrial upgrading policies, or investment-promotion instruments. Australia and New Zealand are classified as liberal-market economies. The Philippines and Indonesia are grouped with them as comparatively more market-oriented, or less state-coordinated, economies in the study sample. This classification is used to examine whether the growth effect of AJCEP differs across economies with different income levels and industrial-policy orientations.

Columns (1) and (2) of Table 8 report the heterogeneity regression results based on income levels. The DID2008 coefficient is 0.428 in the high-income (HI) group and 0.864 in the middle- and lower-income (MLI) group, and both estimates are statistically insignificant. This finding indicates that the subsample estimates do not strongly support income-level heterogeneity, although the coefficient is larger in the MLI group than in the HI group. The result suggests that development-stage differences do not clearly explain the variation in AJCEP's growth effect in this sample.

Therefore, the income-based results do not support H2a. Income level alone may not adequately capture an economy's capacity to translate investment-related institutional improvements into growth. The growth effects of external investment also depend on domestic institutional conditions that influence capital allocation and the productive use of investment (Acemoglu et al., 2005; Okwu

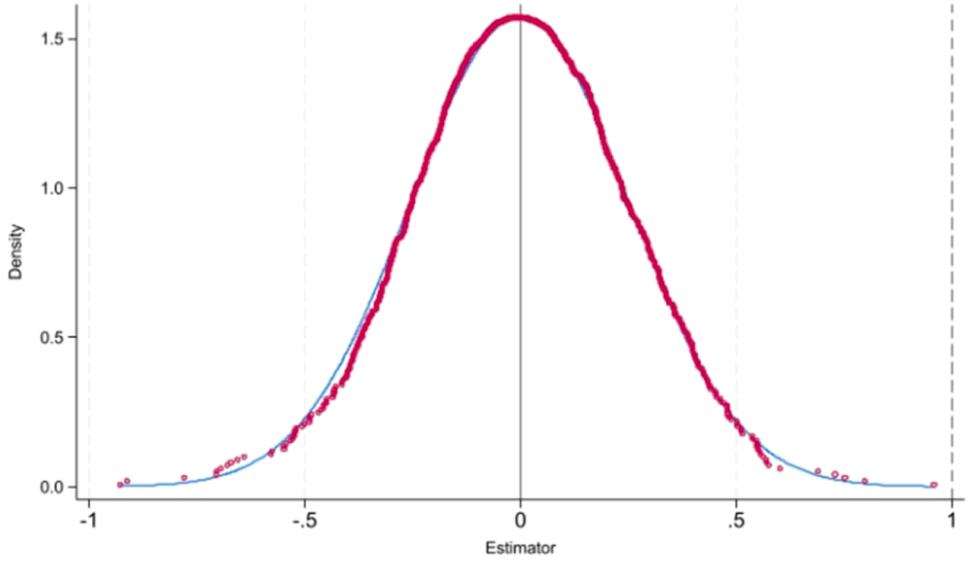


Fig. 2. Placebo Test.

Table 8
Subsample Regression Results for Heterogeneity Analysis.

	(1) HI EP	(2) MLI EP	(3) IIP EP	(4) LME EP
DID2008	0.428	0.864	1.986**	1.951**
	(0.555)	(0.891)	(3.478)	(5.340)
IRC	0.276**	−0.035	0.034	−0.031
	(3.760)	(−0.379)	(0.414)	(−0.436)
GOV	−0.025	0.053	0.050	−0.001
	(−0.494)	(0.813)	(0.709)	(−0.021)
OPEN	−0.004	0.008	0.005	0.033
	(−0.295)	(1.607)	(0.357)	(0.839)
URBAN	−0.133*	−0.169	−0.084	−0.136
	(−2.384)	(−2.600)	(−1.327)	(−1.904)
Constant	16.277***	8.809**	7.814*	10.830
	(10.300)	(6.009)	(2.765)	(1.494)
Observations	144	72	120	96
Number of countryid	6	3	5	4
Adjusted R-squared	0.519	0.541	0.523	0.451
country FE	YES	YES	YES	YES
year FE	YES	YES	YES	YES

Notes: Robust t-statistics are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All specifications include country and year fixed effects. HI denotes high-income economies; MLI denotes middle- and lower-income economies; IIP denotes interventionist industrial-policy economies; LME denotes liberal-market-oriented economies. Significance levels in the subsample regressions are evaluated using the small-sample degrees of freedom associated with the number of countries in each subsample.

et al., 2025). Differences in these conditions across the AJCEP economies may help explain why no clear income-based pattern emerges.

Columns (3) and (4) of Table 8 present heterogeneity results by policy orientation. The results show that, in countries with a stronger interventionist industrial policy (IIP), the DID2008 coefficient is 1.986 and significant at the 5 percent level. In countries where liberal market economies (LMEs) play a dominant role, the coefficient is 1.951 and is significant at the 5 percent level. This finding indicates that the positive growth effect of AJCEP is evident in both interventionist industrial-policy and liberal-market-oriented settings.

The two policy settings can transmit investment-related institutional changes through different channels. In economies with stronger interventionist industrial policy, government coordination may facilitate the transmission of investment-related institutional arrangements through industrial upgrading, infrastructure support, and targeted policy implementation. This is consistent with Li and Wu (2023), who show that indirect government intervention through industrial policy can improve labor investment efficiency by reducing over-hiring and alleviating local government intervention in firm decisions. In more market-oriented economies, property-rights protection, trade freedom, and regulatory efficiency can facilitate foreign investment responses. Tag and Degirmen (2022) find that the rule of law, trade freedom, and regulatory efficiency are positively associated with FDI, with regulatory efficiency remaining significant after controlling for political stability, bilateral investment treaties, and FDI stocks. These institutional conditions help explain why positive AJCEP effects are also observed in the LME group.

Table 8 reports the subsample regression results by income level and industrial-policy orientation. The income-based estimates are positive but statistically insignificant, suggesting that the growth effect of AJCEP is not clearly differentiated by income level in this sample. By contrast, the policy-orientation results show positive and statistically significant DID coefficients in both the IIP and LME groups. This indicates that the growth-enhancing effect of AJCEP is observed across different policy-orientation environments. To examine whether this result is sensitive to the classification of hybrid interventionist cases, Appendix A5 further excludes Malaysia and Thailand and compares the core IIP group with the LME group. The results remain positive and statistically significant in both subsamples, suggesting that the policy-orientation findings are not driven by the inclusion of Malaysia and Thailand in the IIP group.

Overall, the results do not reveal a clear income-based pattern in AJCEP's growth effect. The policy-orientation results show that the effect remains positive under both state-coordinated and market-oriented systems, although the institutional conditions supporting investment differ between the two groups. In IIP economies, policy coordination can facilitate investment and industrial adjustment, whereas in LME economies, property rights protection and efficient market regulation can support foreign investment responses. The similar DID coefficients do not indicate a stronger growth effect under either policy orientation. For H2b, the results therefore suggest that institutional differences are more evident in the way the policy effect is transmitted than in the magnitude of the estimated growth effect.

5. Conclusion and discussion

5.1. Conclusion

This study examines whether investment-related provisions in regional agreements generate identifiable effects on economic growth. Using a quasi-natural experiment based on the implementation of AJCEP and a DID approach, the results show that AJCEP implementation is associated with a significant and economically meaningful improvement in growth among the participating ASEAN economies. The baseline estimate corresponds to an increase of approximately 1.63 percentage points in annual GDP growth relative to the comparison group, compared with an average annual growth rate of 4.64 percent in the full sample. The dynamic estimates further show that the growth effect becomes more visible following implementation. At the same time, the broader empirical analysis confirms the stability of this positive relationship across alternative model specifications and sample adjustments. The heterogeneity results further deepen the interpretation of the overall effect. Income level does not provide a clear dividing line in the growth response, while positive effects are observed in both interventionist industrial-policy and liberal-market settings, with similar estimated magnitudes. Regional investment cooperation can therefore generate measurable growth gains under different domestic policy arrangements. The auxiliary evidence on FDI inflows, production-network trade linkages, and regulatory quality further connects the aggregate growth effect with broader adjustments in regional investment, production organization, and the institutional environment.

These findings contribute to the existing literature at three levels. First, at the literature level, the study connects two strands of research that have largely developed separately. The FDI and growth literature has concentrated on the economic consequences of realized capital inflows, while research on deep regional agreements has focused primarily on trade, investment, multinational production, and participation in global value chains. This study links these two areas by examining whether investment-related institutional commitments embedded in a regional agreement generate measurable aggregate growth effects. It therefore extends the analysis of regional investment policy from its effects on cross-border capital and production to its broader macroeconomic consequences. Second, at the empirical level, AJCEP provides a quasi-experimental setting for evaluating an institutional change that precedes firms' realized investment responses. Treating its implementation as a regional policy shock allows the analysis to examine the growth consequences of the investment policy environment itself, reducing reliance on observed FDI flows as the primary measure of investment conditions. The DID design, together with the dynamic and robustness analyses, provides policy-evaluation evidence that complements the conventional cross-country relationship between FDI and economic growth. Third, at the regional and institutional level, the study shows how a common investment-related regional framework operates within the heterogeneous policy systems of East Asia. The positive effects observed in both IIP and LME economies indicate that regional institutional coordination can be compatible with different domestic approaches to resource allocation and policy implementation. The ASEAN–Japan setting further places these institutional effects within a production system characterized by strong complementarities between investment, trade, and cross-border production. The auxiliary evidence on FDI, production-network linkages, and regulatory quality reinforces this connection and extends the interpretation of regional integration from formal agreement participation to the investment and production environment through which its economic effects are realized.

Several limitations should be acknowledged. First, the analysis is based on a relatively small regional sample, which limits the extent to which the estimated effects can be generalized to other regional agreements and institutional settings. Second, the macro-level reduced-form design does not directly observe firm-level investment decisions, supplier relationships, or adjustments within production networks. The auxiliary evidence, therefore, cannot establish the relative contribution of individual transmission channels. Third, AJCEP represents a comprehensive regional institutional arrangement implemented alongside evolving regional and domestic policies. The empirical design accounts for the major observable policy overlaps, whereas the aggregate framework cannot separately identify the contributions of individual concurrent policy changes or AJCEP provisions. Future research could combine firm-level investment and production-network data with bilateral and sectoral information to more directly identify policy transmission. Firm-level evidence could clarify how regional rules affect investment location, local supplier relationships, and production upgrading. At the same time, bilateral analysis could examine how Japan–ASEAN investment and production linkages respond to institutional change. Comparative analysis of regional agreements with varying investment provisions could further identify which institutional features are most effective at generating sustained gains in investment, productivity, and growth.

5.2. Discussion

The findings of this study provide new empirical evidence for understanding the relationship between changes in investment policy and economic growth. While existing literature has predominantly focused on the economic effects of FDI, relatively little attention has been paid to the macroeconomic consequences of investment policies and their associated institutional arrangements (Ashraf et al., 2024). This study shifts the focus from FDI flows to the role of investment policy and the institutional environment. Using the AJCEP implementation to construct a quasi-natural experiment provides a novel empirical framework for analyzing how investment policy shocks influence economic growth. The positive growth effect is consistent with Ma (2022), who finds that broader economic integration under the Belt and Road Initiative improves economic performance. The present findings extend this evidence by showing that measurable growth gains are also associated with investment-related institutional changes within a specific regional agreement. Medvedev (2012) finds that preferential trade agreements are associated with higher net FDI inflows, particularly in developing economies and during the expansion of deep integration agreements. This evidence suggests that regional agreements can influence both investment decisions and trade. In the AJCEP context, greater policy predictability can reduce uncertainty surrounding long-term capital commitments. Inada and Jinji (2024) show that uncertainty regarding investment treatment discourages FDI by Japanese

multinational enterprises. More predictable investment rules can therefore strengthen firms' willingness to undertake long-term investment, thereby expanding productive capacity and supporting economic growth.

The results also help clarify the mixed findings in the literature regarding the relationship between FDI and economic growth. Previous research shows that FDI's contribution to economic growth depends on host-country institutional conditions (Li et al., 2023). The heterogeneity results provide further evidence on this issue. The income-based estimates do not support H2a, while the AJCEP effect remains positive and significant in both IIP and LME economies. The estimated effects are also similar in magnitude, which does not indicate a systematically stronger growth response under either policy orientation. The main distinction, therefore, lies in the way the policy effect is transmitted. In IIP economies, government coordination and industrial policy can facilitate investment adjustment and industrial upgrading. Li and Wu (2023) show that indirect industrial policy intervention can improve the efficiency of firm-level labor investment. In more market-oriented economies, the rule of law, trade freedom, and regulatory efficiency are associated with higher FDI (Tag and Degirmen, 2022). These results suggest that different domestic institutional arrangements can facilitate the transmission of investment-related policy changes, even when the resulting growth effects are of similar magnitude.

This study also contributes to the literature on regional economic integration. Regional agreements can influence economic performance by changing investment incentives and the organization of cross-border production. In East Asia, where cross-border production networks have long been central to regional integration, trade, and FDI policies can influence multinational production and the organization of regional production networks (Chen et al., 2026). Recent evidence also helps explain how these changes can affect productivity. Yang et al. (2025) find that stronger RTA-network integration improves countries' positions in global value chains and identify total factor productivity as an important transmission mechanism. Wang and Yang (2025) provide complementary firm-level evidence that deeper RTA-network embeddedness improves productivity through trade expansion and innovation. These findings show how regional production integration can contribute to economic performance through productivity improvement, trade expansion, and technological upgrading. The auxiliary evidence in Appendix A6 is consistent with this interpretation. AJCEP is associated with higher FDI inflows, stronger production-network trade linkages, and improved regulatory quality among the treated economies. These supplementary results further suggest that investment-related regional agreements may affect growth through a broader institutional and production-network environment, rather than through capital inflows alone (de Soyres et al., 2021; Mattoo et al., 2020; Yue et al., 2023).

5.3. Policy implications

The empirical findings offer several policy implications for ASEAN member states seeking to use regional integration more effectively to support economic growth. The recommendations are particularly relevant to Indonesia, the Philippines, Vietnam, Thailand, and Malaysia, as well as ASEAN regional institutions and Japanese counterparts involved in ASEAN–Japan economic cooperation.

First, national investment authorities should place greater emphasis on the predictability and consistency of investment policy implementation. In Indonesia, changes in sector-specific foreign investment restrictions should be accompanied by clear transition arrangements and timely communication with affected firms. Sudden or ambiguous regulatory changes may disrupt capital allocation and weaken firms' willingness to undertake long-term investment. In the Philippines, investment authorities should improve the transparency of application requirements, approval procedures, and processing timelines. Practical measures include digital one-stop investment services, regularly updated regulatory databases, clear notification of policy changes, and accessible investor grievance mechanisms. These measures are also important for Japanese investors, whose long-term investment decisions depend heavily on the stability and clarity of investment treatment. More predictable implementation can therefore strengthen the credibility of AJCEP commitments and encourage sustained investment.

Second, policies aimed at converting FDI into domestic productivity gains should reflect differences in national production structures and positions within regional production networks. In Vietnam, the priority should be to strengthen linkages between foreign manufacturers and domestic suppliers. Investment and industrial agencies can support supplier certification, technical testing, workforce training, and closer connections between domestic suppliers and multinational enterprises, including Japanese firms operating within regional production networks. For Thailand, greater emphasis should be placed on downstream production, product development, and the upgrading of domestic firms beyond component manufacturing. Indonesia should strengthen local supplier capabilities, promote industrial diversification, and encourage greater participation in higher-value stages of regional production. Malaysia, given its deeper integration into regional production networks, should focus more on research and development, product design, advanced technical skills, and stronger innovation linkages between multinational and domestic firms.

Third, regional institutions should focus on reducing implementation differences across overlapping regional frameworks. The ASEAN Coordinating Committee on Investment and the ASEAN Secretariat, together with relevant Japanese counterparts, can improve administrative consistency between AJCEP and RCEP. Practical measures could include a common digital repository of investment rules and approval procedures, regular reviews of implementation differences, a regional investor inquiry and referral mechanism, and joint supplier-matching and technical-certification programs. Coordination can also extend to digital customs procedures and the mutual recognition of technical standards when production activities cross national borders. Japan can contribute through supplier development, technical training, certification, and closer links between Japanese multinational enterprises and qualified ASEAN suppliers. The central objective is to reduce the administrative and coordination costs faced by firms operating across multiple member economies and to improve the efficiency of regional production networks.

Finally, the heterogeneity results suggest that policy design should not rely primarily on income classification. The estimated AJCEP effect remains positive in both the IIP and LME settings, indicating that the benefits of regional integration are not confined to

any particular income group. Policy implementation should therefore give greater attention to differences in domestic institutional structures and policy capacity. Economies with stronger state coordination can use industrial and investment agencies to organize supplier development, skills programs, and post-investment services. More market-oriented systems can place greater emphasis on transparent regulation, predictable entry conditions, and private-sector participation. Policy implementation should also be monitored using measurable indicators, including investment approval times, investor grievance resolution rates, local sourcing by foreign firms, the number of certified domestic suppliers, and participation in training or research programs. National investment agencies can report these indicators regularly, while the ASEAN Coordinating Committee on Investment can use them to assess progress in regional implementation.

Ethical approval

This study does not contain any studies with human participants performed by any of the authors.

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CRediT authorship contribution statement

Shasha Wu: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Yang Liu:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis. **Shun Li:** Writing – original draft, Data curation. **Fumitaka Furuoka:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A1

Table A1

Table A1
Baseline Regression Results.

VARIABLES	EP
DID2008	1.630***
	(3.999)
IRC	0.034
	(0.564)
GOV	0.028
	(0.865)
OPEN	0.004
	(0.497)
URBAN	−0.160***
	(−3.518)
Constant	12.456***
	(8.742)
Observations	216
Number of countries	9
Adjusted R-squared	0.527
Country FE	YES
Year FE	YES
SE type	Country clustered
Wild bootstrap p-value	0.0173
Wild bootstrap 95% CI	[0.5678, 2.7190]

Notes: Country-clustered t-statistics are reported in parentheses. The wild cluster bootstrap test is based on 9999 replications with clustering at the country level, and the reported interval is the 95% confidence interval;*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix A2

Fig. A2

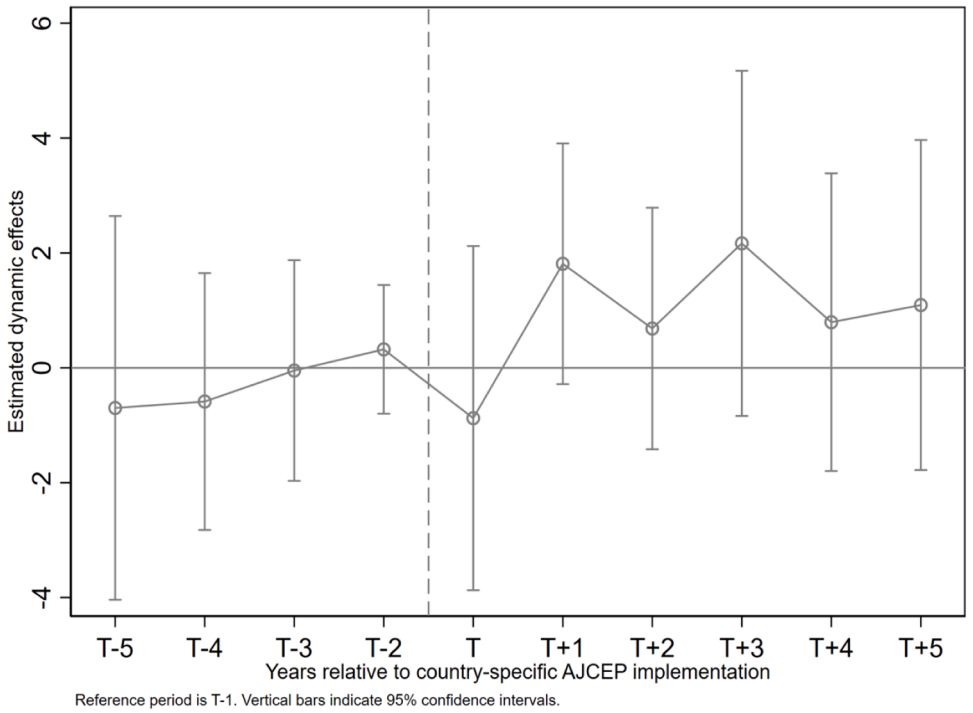


Fig. A2. Parallel Trend Graph.

Appendix A3

Table A3

Table A3
Control-Group Composition Robustness Check.

VARIABLES	EP
DID2008	2.325***
	(3.588)
IRC	0.030
	(0.468)
GOV	0.064**
	(1.994)
OPEN	0.001
	(0.169)
URBAN	−0.119**
	(−2.589)
Constant	8.298***
	(2.880)
Observations	168
Number of countryid	7
Adjusted R-squared	0.566
Country FE	YES
Year FE	YES
Excluded countries	Australia and New Zealand

Notes: t-statistics are reported in parentheses. Australia and New Zealand are excluded from the control group; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix A4

Table A4

Table A4
Treatment-Group Composition Robustness Check.

VARIABLES	EP
DID2008	1.140**
	(2.091)
IRC	0.049
	(0.733)
GOV	0.035
	(1.125)
OPEN	0.014*
	(1.821)
URBAN	−0.115***
	(−2.651)
Constant	8.789***
	(2.888)
Observations	264
Number of countryid_ext	11
Adjusted R-squared	0.402
Country FE	YES
Year FE	YES
Treatment group	Expanded

Notes: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.
Singapore and Brunei are added to the treatment group to examine whether the baseline result is sensitive to their exclusion from the main treatment sample. The estimation uses the preferred baseline specification with country and year fixed effects.

Appendix A5

A threshold-sensitivity analysis of the policy-orientation classification used in the heterogeneity analysis is reported in Table A5. Under the baseline IIP/LME framework, China, Korea, Malaysia, Thailand, and Vietnam are classified as interventionist industrial-policy economies. However, Malaysia and Thailand may be more appropriately characterized as hybrid interventionist cases, as their pre-AJCEP policy environments combined market-oriented elements with industrial planning, investment promotion, and industrial upgrading policies.

Table A5
Robustness Check for Policy-Orientation Classification.

VARIABLES	(1) Core IIP	(2) LME
DID2008	1.646*	1.951**
	(3.118)	(5.340)
IRC	−0.055	−0.031
	(−0.484)	(−0.436)
GOV	−0.107	−0.001
	(−1.127)	(−0.021)
OPEN	0.024	0.033
	(1.749)	(0.839)
URBAN	0.014	−0.136
	(0.313)	(−1.904)
Constant	11.774*	10.830
	(3.862)	(1.494)
Observations	72	96
Number of countryid	3	4
Adjusted R-squared	0.477	0.451
Country FE	YES	YES
Year FE	YES	YES

Notes: t-statistics are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All specifications include country and year fixed effects. Significance levels are evaluated using the small-sample degrees of freedom associated with the number of countries in each subsample.

To examine whether the policy-orientation results depend on the inclusion of these hybrid cases, Table A5 excludes Malaysia and Thailand and compares the core IIP group with the LME group. The core IIP group includes China, Korea, and Vietnam, while the LME group includes Australia, New Zealand, the Philippines, and Indonesia. The DID coefficient remains positive and statistically significant in both subsamples. Specifically, the DID coefficient is 1.646 in the core IIP group and 1.951 in the LME group. These results suggest that the positive policy-orientation findings are not driven by the inclusion of Malaysia and Thailand in the IIP group and remain observable under a stricter definition of interventionist industrial-policy economies.

Appendix A6

Using three macro-level proxy variables associated with the theoretical channels discussed in the manuscript, Table A6 presents auxiliary channel-level evidence. The estimations are based on a DID framework with country and year fixed effects.

Column (1) uses FDI inflows as the dependent variable to capture the investment-related channel. The DID coefficient is positive and statistically significant at the 1 percent level, with a coefficient of 1.131 and a t-statistic of 2.854. This suggests that implementing AJCEP is associated with higher FDI inflows in the treated economies. The increase in FDI is consistent with an investment expansion response to changes in the regional investment environment. Column (2) uses the share of manufactured imports in merchandise imports as a proxy for production-network trade linkages. This variable captures broad trade integration related to production networks, rather than direct intermediate-goods trade. The DID coefficient is 7.639 and statistically significant at the 1 percent level, suggesting that AJCEP implementation is associated with stronger production-network trade linkages. This pattern indicates that the policy response also involves deeper integration into manufacturing-based regional production networks. Column (3) uses the WGI regulatory-quality percentile rank as the institutional proxy. The DID coefficient is positive and statistically significant at the 1 percent level, with a coefficient of 3.734 and a t-statistic of 2.685. This indicates that the implementation of AJCEP is associated with improved regulatory quality in the treated economies. A stronger regulatory environment can reduce institutional frictions and improve the predictability of long-term investment decisions.

Table A6
Auxiliary Channel Proxy Results.

VARIABLES	(1) FDI inflows	(2) Trade proxy	(3) Regulatory quality
DID	1.131***	7.639***	3.734***
	(2.854)	(7.479)	(2.685)
IRC	0.098**	−0.361***	−0.273
	(2.090)	(−2.905)	(−1.634)
GOV	0.014	0.074	0.504***
	(0.649)	(1.243)	(6.466)
OPEN	0.003	0.099***	−0.008
	(0.439)	(5.877)	(−0.326)
URBAN	−0.125***	−0.522***	−0.511***
	(−4.014)	(−6.396)	(−4.395)
Constant	7.114***	90.191***	58.040***
	(3.487)	(16.950)	(7.913)
Observations	216	212	198
Number of country_num	9	9	9
Adjusted R-squared	0.112	0.651	0.402
Country FE	YES	YES	YES
Year FE	YES	YES	YES

Notes: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

These results are consistent with the interpretation that investment-related regional agreements may support economic growth by expanding investment, strengthening production-network linkages, and improving regulatory environments.

Data availability

The cleaned panel dataset and replication code supporting the findings of this study have been deposited in the *Science Data Bank* (https://doi.org/10.57760/sciencedb.40700).

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